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CONTACT INFORMATION

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RESEARCH FIELDS

Environmental and Energy Economics
Empirical Industrial Organization

EDUCATION

University of Maryland, College Park, 2021–Present
Ph.D. Candidate in Agricultural and Resource Economics
Expected May 2027

University of Chicago, 2021
M.A. in Public Policy; Certificate in Research Methods

Jilin University, 2019
B.A. in Economics

REFERENCES

Joshua Linn (co-advisor)
Professor
University of Maryland
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Chenyu Yang (co-advisor)
Assistant Professor
University of Maryland
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Louis Preonas
Associate Professor
University of Maryland
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Maureen L. Cropper
Distinguished University Professor
University of Maryland
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WORKING PAPERS

"Foreign Entry, Local Sourcing, and Domestic Welfare: Evidence from Tesla's Manufacturing in China" *(Job Market Paper)*

Abstract: Products from a leading foreign manufacturer can hurt domestic rivals through competition, yet its local production and sourcing can help them through spillovers that lower their production costs and fixed costs of introducing new products. I estimate a model of consumer demand, oligopoly pricing, and product introduction to study these opposing forces and their welfare effects in China's passenger vehicle market, where Tesla began producing electric vehicles (EVs) in Shanghai and sourcing their components locally in 2019. The marginal costs of rival EVs sharing Tesla's core-part suppliers fell more than those of other rival EVs. The fixed costs of introducing battery-electric models similar in size to Tesla's locally produced models also fell relative to those of smaller battery-electric models. Counterfactual analysis shows that competition from Tesla's Shanghai-built models discourages rivals from introducing new products, while these cost reductions encourage them. On net, rivals' new product offerings rise by about 32 percent relative to a market without Tesla's local production and sourcing. Domestic welfare rises, and consumers receive most of the gain. Allowing rivals to adjust their offerings raises consumers' gains by about 27 percent. Domestic producer gains accrue to EV start-ups because their models share more suppliers with Tesla and are closer to its models in size. Established Chinese automakers lose.

"Geographic Resolution in Environmental Policy: EPA's Shift from Regions to Counties Under the Clean Air Act," with Maureen Cropper, Nicholas Muller, and Yongjoon Park. *NBER Working Paper 33412.*

Abstract: Prior to 1978 EPA regulated nonattainment with the Clean Air Act using Air Quality Control Regions (AQCRs), switching to counties in 1978. We find that EPA's official AQCR nonattainment designation for total suspended particulate matter (TSP) in 1972 had no causal effect on ambient TSP. Nor do we detect evidence that being in nonattainment in the AQCR regime affected employment in pollution intensive manufacturing industries. In contrast, when nonattainment is confined to smaller areas—counties in 1978 or counties with monitor violations in 1972, we find evidence that nonattainment decreased both ambient TSP and manufacturing employment.

"Unintended Consequences of Phytosanitary Regulation: Smuggling and Exotic Pest Risk," with Rebecca Epanchin-Niell.

Abstract: Phytosanitary regulations reduce pest risk from legal trade but may encourage unauthorized exports that bypass biosecurity safeguards. We develop a bioeconomic model in which producers allocate output among domestic sales, certified exports, and smuggling. We examine how certification requirements, enforcement, and penalties affect total expected pest risk and social welfare, and how these outcomes depend on prices, costs, firm size, and industry structure. We find that stricter phytosanitary requirements can unintentionally increase overall pest risk by shifting trade toward unauthorized pathways. The balance between intended risk reductions and increased smuggling depends on enforcement, price differences between domestic and foreign markets, and industry structure. Differences in production costs and expected penalties across firms further shape these responses: under uniform regulations, smaller producers may smuggle a larger share of exports and abandon certification at lower stringency. These findings highlight the importance of accounting for evasion incentives when evaluating phytosanitary policy. Coordinating certification requirements with enforcement can help reduce pest risk and improve welfare.

WORKS IN PROGRESS

"Home Charging Access, Indirect Network Effects, and Optimal Subsidy Strategy: Evidence from China's EV Market," with Joshua Linn, Lunyu Xie, and Tianhua Zou.

"GenAI and the Geography of Skill-Biased Deployment in the Energy Transition," with Victoria Wenxin Xie and Wu Zhu.

"IRA Subsidies for EVs, Import Tariffs, and Domestic Industry," with Joshua Linn and Louis Preonas.

"Remote Work, Travel Behavior, and Public Transit Adaptation," with Yixin Yang.

PRESENTATIONS

2027: ASSA, AEA (scheduled, January).
2026: Southern Economic Association Annual Meeting (scheduled, November).
2025: AAEA; AERE@WEAI; UMD AREC 7th Annual Cornerstone Event.
2024: UMD AREC Research Workshop; UMD Annual AREC Poster Presentations.

TEACHING

Teaching Assistant, University of Maryland, College Park
Introduction to Economics and the Environment (undergraduate), 2024
Applied Econometrics II (Ph.D. level), 2023
World Hunger, Population, and Food Supplies (undergraduate), 2021, 2022

FELLOWSHIPS AND AWARDS	2026–2027: Ann G. Wylie Dissertation Fellowship, University of Maryland 2025: Bruce and Mary Ann Gardner Dissertation Award, University of Maryland 2025: Goldhaber Travel Grant, Graduate School, University of Maryland 2021–2026: Graduate Assistantship, University of Maryland 2021: Graduate Student Fellowship, University of Maryland 2019: Outstanding Undergraduate Thesis (top 5%), Jilin University 2017–2018: College Excellent Student, Jilin University 2017–2018: Second Class Scholarship, Jilin University
PROFESSIONAL EXPERIENCE	**Research Consultant, Resources for the Future, 2024** *Retrospective Analysis of Environmental Regulation* **Research Assistant, University of Maryland, College Park** Joshua Linn and Louis Preonas, 2026 Rebecca Epanchin-Niell, Aug. 2024–Present Louis Preonas, Sep. 2023–Jan. 2024 Anna Alberini, Feb.–Jul. 2022
CODING SKILLS	Python, Stata, R, MATLAB, LaTeX
LANGUAGES	Mandarin Chinese (native); English (full professional proficiency)